QUALCOMM

pdQ SDK Lab
PalmSource™ 99
Session 188



Clarence Wong

Lab Overview

- Developer Zone and Developer Basics
- High Level Review of APIs
- Demos
- Real World Solution: Stock Quote Alert app
- Q&A



pdQ Development Zone

- www.qualcomm.com/phones/products/pdq_phone
- SDK available online
- pdQ mailing lists
- Developer email questions pdqdeveloper@qualcomm.com



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pdQ Architecture

Gajinder Vij



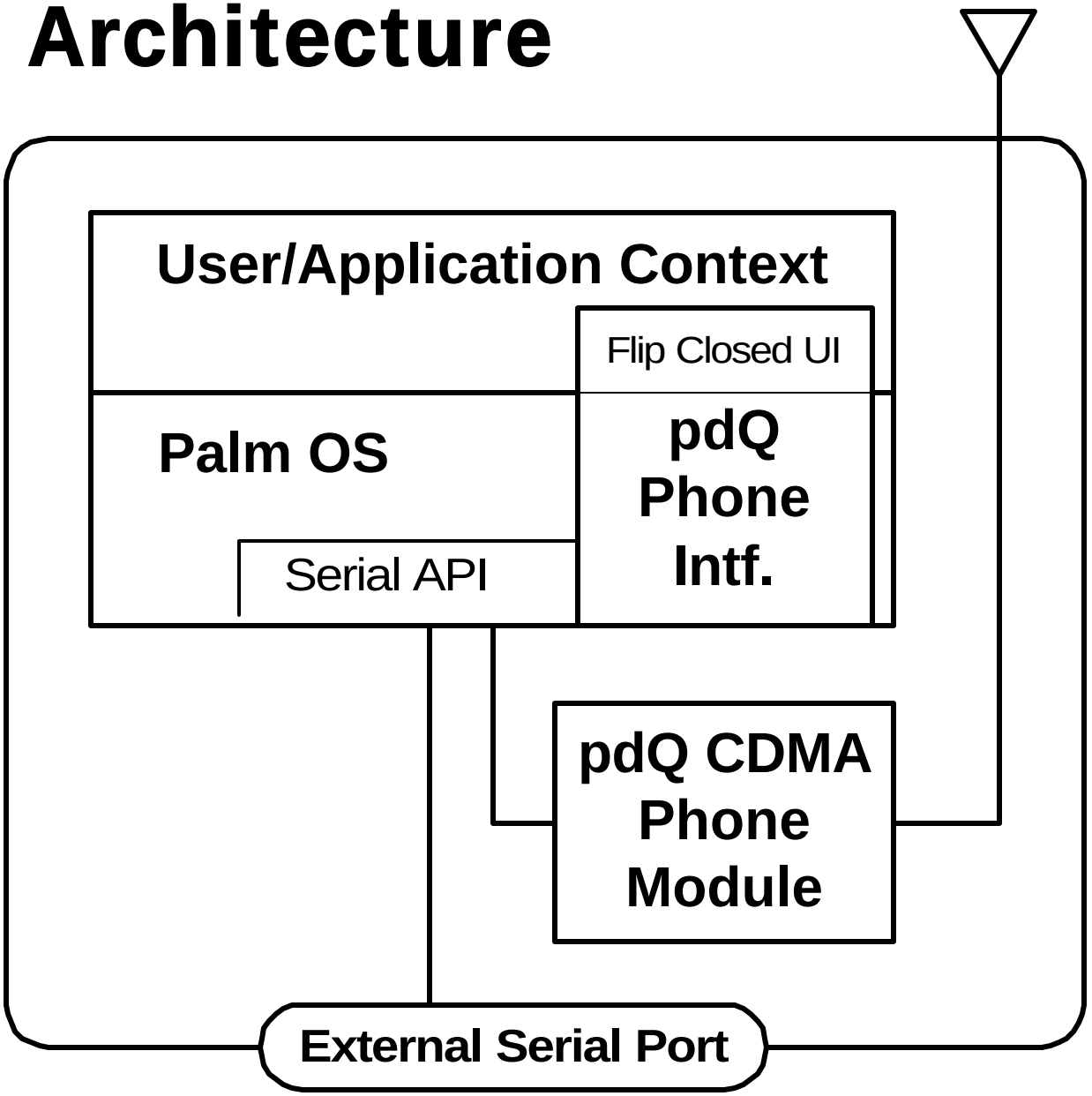
pdQ Architecture Overview

- pdQ is based on Palm OS® 3.02
- Equivalent to a Palm III™ Device
- Phone Intf. runs as an extension to Palm OS
- Compatible with apps using Palm OS API (e.g., Palm Platinum certified apps)
- Non-Palm API calls will most likely fail (i.e. Direct screen access)



pdQ Architecture





pdQ Memory Details

- Phone Intf uses part of available Dynamic Heap
- 62k Dynamic Heap available for Non-Data apps vs 68k for Palm III
- 30k Dynamic heap available for Data apps (using Netlib) vs 36k for Palm III



pdQ Mobile-Originated Data



pdQ Mobile-Originated Data

- Make TCP/IP Mobile-originated data calls using Palm OS Netlib
- What's different? No difference except Memory constraints
- Consider pdQ equal to a Palm III device with a modem
- Most existing NetLib applications will work
- How to connect? Open a socket, pdQ does the rest



pdQ Data—Hints

- Use standard socket programming
- Recommend using non-blocking I/O
- Call SysHandleEvent() occasionally to update signal indicators, call timer, etc.
- Optionally use pdQ Signal Manager to handle terminated data calls gracefully
- NetLib memory constraints



pdQ Data—Setup

- Data implementation is carrier specific (QNC/Async). Connection handled through Network Prefs panel
- Carrier configures pdQ for data. Network Prefs settings pre-configured for carrier's network
- Use pdQbrowser to verify pdQ configuration and carrier support for data



pdQ Data

- All Netlib (TCP/IP) applications - to enable Wireless Data, user chooses ‘QUALCOMM Wireless’ as the modem selection from the Modem Prefs panel (this is default on pdQ)

- Non-TCP/IP apps requires a change to support a wireless connection. Additional info provided with the PalmSource 99 pdQ Lab Sample Code



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pdQ SDK Overview

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pdQ SDK Overview

- Telephony API: Phone control
- Alert API: Get users attention
- Registry API: Custom and standard URL schemes
- Signal API: Inter-application communication



pdQ Telephony API

- Provides control of voice calls
- Answer voice calls
- End voice calls
- Make voice calls
  - Handles digits [1-619-555-1212]
  - Handles alpha [1-800-FLOWERS]
  - Handles pauses [voice mail log in and macros]



pdQ Telephony Demo

Make a call…



pdQ Alert API

- Used to notify user or get attention
- Turn back light on and off
- Control LED
  - Red, green, orange
  - Flash rate control
- *(Note: use Palm OS ® software to make sounds)*



pdQ Alert Demo

SimonSez…



pdQ URL Registry API

- Manage and dispatch URLs
- Built in schemes include
  - http: sends information to pdQbrowser
  - mailto: sends information to pdQmail
  - tel: sends information to the dialer
- Create custom application scheme (stock-graph:)



pdQ URL Registry Demo

Dispatch a mailto: URL...



pdQ Signal API

- Allows apps to be notified when events happen
- Built in signals for common phone functions
- Developer can add custom signals for other applications to use
- Uses standard Palm launch codes to pass information



pdQ Signal API—Built In

- Idle, incoming
- Lost, missed, failed
- Calling, In conversation, ended
- Dial paused, caller ID, DTMF
- SMS, voice mail, text messages



pdQ Signal Demo

Minute Counter

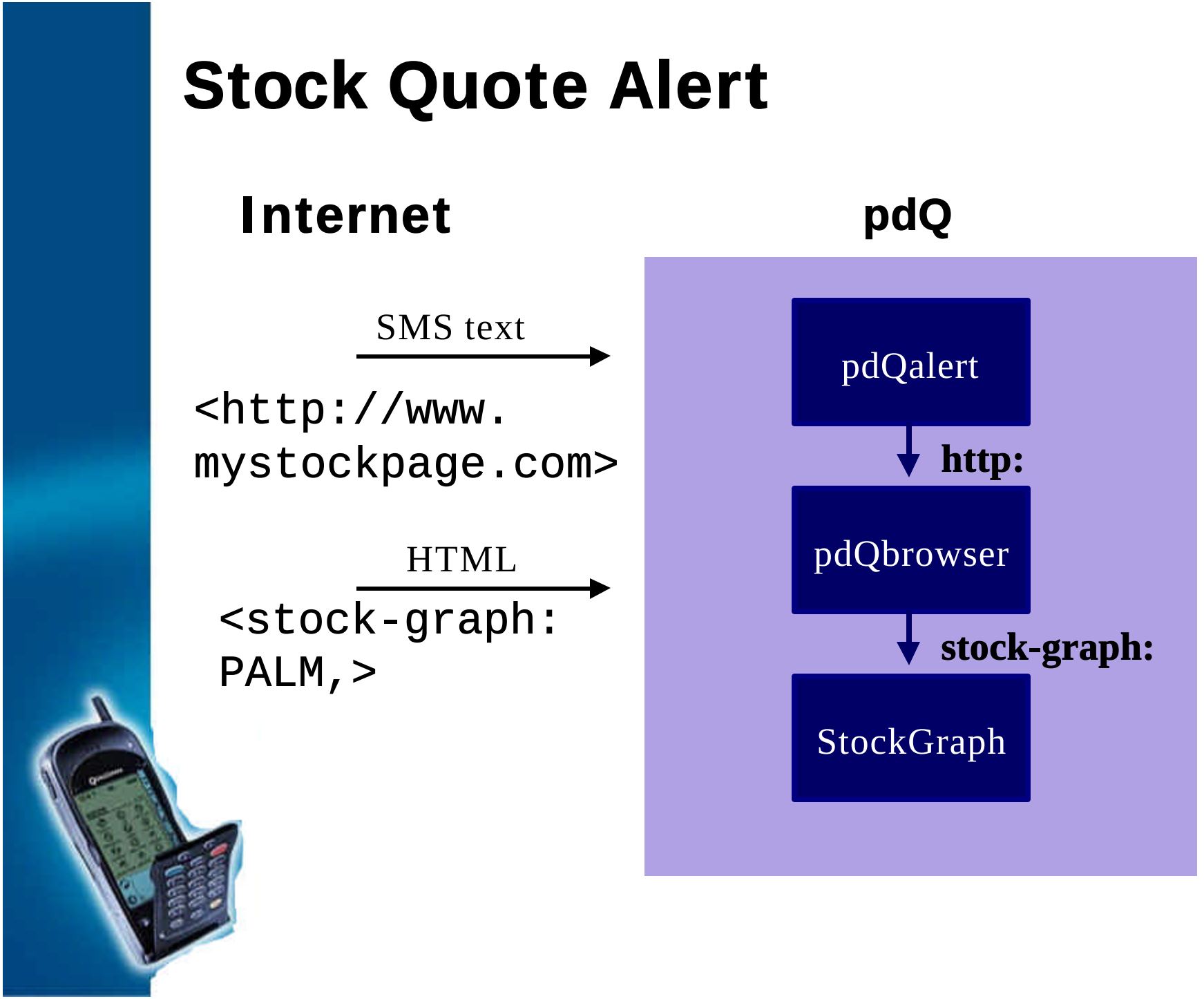


A Real World Solution

- Write a stock quote alert app that:
  - Alerts user that stock has jumped
  - Launches pdQbrowser to get data
  - Launches custom viewer app to process data
  - Allows app registration via e-mail, web, or phone



Stock Quote Alert



Summary

- SDK available on developer zone
- Consider device equivalent to Palm III and modem
- TCP/IP data is available via NetLib
- Four Basic APIs to access phone functionality



Q&A

www.qualcomm.com

